Assignment dql

```
mysql> show databases;
+--------------------------+
| Database                 |
+--------------------------+
| 03_march_sql             |
| 10th_march_sql           |
| asg_ddl_2                |
| assignment_ddl_1         |
| assignment_ddl_2         |
| assignment_dml           |
| for_practice_constraints |
| information_schema       |
| mysql                    |
| performance_schema       |
| sakila                   |
| sqltest1setb             |
| sys                      |
| world                    |
+--------------------------+
14 rows in set (0.06 sec)

mysql> create database assignment_dql_select;
Query OK, 1 row affected (0.08 sec)

mysql> show databases;
+--------------------------+
| Database                 |
+--------------------------+
| 03_march_sql             |
| 10th_march_sql           |
```

```
| asg_ddl_2              |
| assignment_ddl_1       |
| assignment_ddl_2       |
| assignment_dml         |
| assignment_dql_select  |
| for_practice_constraints |
| information_schema     |
| mysql                  |
| performance_schema     |
| sakila                 |
| sqltest1setb           |
| sys                    |
| world                  |
+--------------------------+
15 rows in set (0.00 sec)

mysql> use assignment_dql_select;
Database changed
mysql> show tables;
Empty set (0.00 sec)

mysql> CREATE TABLE Employees (
    -> EMPLOYEE_ID INT PRIMARY KEY,
    -> FIRST_NAME VARCHAR(50),
    -> LAST_NAME VARCHAR(50),
    -> EMAIL VARCHAR(50),
    -> PHONE_NUMBER VARCHAR(20),
    -> HIRE_DATE DATE,
    -> JOB_ID VARCHAR(20),
    -> SALARY DECIMAL(10,2),
    -> COMMISSION_PCT DECIMAL(5,2),
```

```
    -> MANAGER_ID INT,
    -> DEPARTMENT_ID INT
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Employees VALUES
    ->  (100, 'Steven', 'King', 'SKING', '515.123.4567', '1987-06-17', 'AD_PRES', 24000.00, 0.00, 200, 10),
    ->  (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', '1987-06-18', 'AD_VP', 17000.00, 0.00,
200, 10),
    ->  (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', '1987-06-19', 'AD_VP', 17000.00, 0.00, 200,
10),
    ->  (103, 'Alexander', 'Hunold', 'AHUNOLD', '590.423.4567', '1987-06-20', 'IT_PROG', 9000.00, 0.00,
103, 60),
    ->  (104, 'Bruce', 'Ernst', 'BERNST', '590.423.4568', '1987-06-21', 'IT_PROG', 6000.00, 0.00, 103, 60),
    ->  (105, 'David', 'Austin', 'DAUSTIN', '590.423.4569', '1987-06-22', 'IT_PROG', 4800.00, 0.00, 103,
60),
    ->  (106, 'Valli', 'Pataballa', 'VPATABAL', '590.423.4560', '1987-06-23', 'IT_PROG', 4800.00, 0.00, 103,
60),
    ->  (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.423.5567', '1987-06-24', 'IT_PROG', 4200.00, 0.00,
114, 30),
    ->  (108, 'Nancy', 'Greenberg', 'NGREENBE', '515.124.4569', '1987-06-25', 'SA_MAN', 12000.00,
0.00, 145, 80),
    ->  (109, 'Daniel', 'Faviet', 'DFAVIET', '515.124.4169', '1987-06-26', 'SA_MAN', 9000.00, 0.00, 145,
80);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> CREATE TABLE Department (
    -> Department_ID INT PRIMARY KEY,
    -> Department_Name VARCHAR(50),
    -> Manager_ID double,
    -> Location_ID double
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Department (Department_ID, Department_Name, Manager_ID, Location_ID)
VALUES
    ->  (10, 'Administration', 200, 1700),
    ->  (20, 'Marketing', 201, 1800),
    ->  (30, 'Purchasing', 114, 1700),
    ->  (40, 'Human Resources', 203, 2400),
    ->  (50, 'Shipping', 121, 1500),
    ->  (60, 'IT', 103, 1400),
    ->  (70, 'Public Relations', 204, 2700),
    ->  (80, 'Sales', 145, 2500);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Department (Department_ID, Department_Name, Manager_ID, Location_ID)
    -> VALUES
    ->  (90, 'Finance', 205, 2700),
    ->  (100, 'Legal', 206, 2200),
    ->  (110, 'Customer Service', 207, 2300);
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> create table jobs(
    ->     job_id varchar(20) primary key,
    ->   job_title varchar(50) not null,
    -> min_salary double,
    ->     max_salary double
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> > INSERT INTO Jobs (JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY) VALUES
```

-> ('AD_PRES', 'President', 20000, 40000),

-> ('AD_VP', 'Administration Vice President', 15000, 30000),

-> ('AD_ASST', 'Administration Assistant', 3000, 6000),

-> ('FI_MGR', 'Finance Manager', 8200, 16000),

-> ('FI_ACCOUNT', 'Accountant', 4200, 9000),

-> ('AC_MGR', 'Accounting Manager', 8200, 16000),

-> ('AC_ACCOUNT', 'Public Accountant', 4200, 9000),

-> ('SA_MAN', 'Sales Manager', 10000, 20000);

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '> INSERT INTO Jobs (JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY) VALUES

('AD_PRES' at line 1

mysql> > INSERT INTO Jobs (JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY) VALUES

-> ('AD_PRES', 'President', 20000, 40000),

-> ('AD_VP', 'Administration Vice President', 15000, 30000),

-> ('AD_ASST', 'Administration Assistant', 3000, 6000),

-> ('FI_MGR', 'Finance Manager', 8200, 16000),

-> ('FI_ACCOUNT', 'Accountant', 4200, 9000),

-> ('AC_MGR', 'Accounting Manager', 8200, 16000),

-> ('AC_ACCOUNT', 'Public Accountant', 4200, 9000),

-> ('SA_MAN', 'Sales Manager', 10000, 20000);

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '> INSERT INTO Jobs (JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY) VALUES

('AD_PRES' at line 1

mysql> INSERT INTO jobs (JOB_ID, JOB_TITLE, MIN_SALARY, MAX_SALARY) VALUES

-> ('AD_PRES', 'President', 20000, 40000),

-> ('AD_VP', 'Administration Vice President', 15000, 30000),

-> ('AD_ASST', 'Administration Assistant', 3000, 6000),

-> ('FI_MGR', 'Finance Manager', 8200, 16000),

-> ('FI_ACCOUNT', 'Accountant', 4200, 9000),

-> ('AC_MGR', 'Accounting Manager', 8200, 16000),

```
    ->    ('AC_ACCOUNT', 'Public Accountant', 4200, 9000),
    ->  ('SA_MAN', 'Sales Manager', 10000, 20000);
Query OK, 8 rows affected (0.06 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> update employees
    -> set salary = salary * 1.10;
Query OK, 10 rows affected (0.06 sec)
Rows matched: 10  Changed: 10  Warnings: 0

mysql> UPDATE Employees
    -> SET EMAIL = 'not available'
    -> WHERE DEPARTMENT_ID = 80 AND COMMISSION_PCT < 0.002;
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> update employees
    -> set email = "not available"
    -> where department_name = "purchasing";
ERROR 1054 (42S22): Unknown column 'department_name' in 'where clause'
mysql> update employees
    ->   set email = "not available"
    -> where department_id = 30;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
```

```
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-------------+
| 100 | Steven | King | SKING | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 | 0.00 | 200 | 10 |
| 101 | Neena | Kochhar | NKOCHHAR | 515.123.4568 | 1987-06-18 | AD_VP | 18700.00 | 0.00 | 200 | 10 |
| 102 | Lex | De Haan | LDEHAAN | 515.123.4569 | 1987-06-19 | AD_VP | 18700.00 | 0.00 | 200 | 10 |
| 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 | 0.00 | 103 | 60 |
| 104 | Bruce | Ernst | BERNST | 590.423.4568 | 1987-06-21 | IT_PROG | 6600.00 | 0.00 | 103 | 60 |
| 105 | David | Austin | DAUSTIN | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 | 0.00 | 103 | 60 |
| 106 | Valli | Pataballa | VPATABAL | 590.423.4560 | 1987-06-23 | IT_PROG | 5280.00 | 0.00 | 103 | 60 |
| 107 | Diana | Lorentz | not available | 590.423.5567 | 1987-06-24 | IT_PROG | 4620.00 | 0.00 | 114 | 30 |
| 108 | Nancy | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN | 13200.00 | 0.00 | 145 | 80 |
| 109 | Daniel | Faviet | not available | 515.124.4169 | 1987-06-26 | SA_MAN | 9900.00 | 0.00 | 145 | 80 |
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-------------+

10 rows in set (0.00 sec)


mysql> select * from department;

+---------------+-----------------+------------+-------------+

| Department_ID | Department_Name | Manager_ID | Location_ID |

+---------------+-----------------+------------+-------------+

| 10 | Administration | 200 | 1700 |

| 20 | Marketing | 201 | 1800 |

| 30 | Purchasing | 114 | 1700 |

| 40 | Human Resources | 203 | 2400 |

| 50 | Shipping | 121 | 1500 |
```

```
|          60 | IT              |       103 |       1400 |
|          70 | Public Relations |       204 |       2700 |
|          80 | Sales           |       145 |       2500 |
|          90 | Finance         |       205 |       2700 |
|         100 | Legal           |       206 |       2200 |
|         110 | Customer Service |       207 |       2300 |
+---------------+------------------+------------+-------------+
11 rows in set (0.05 sec)

mysql> select * from jobs;
+------------+------------------------------+------------+------------+
| job_id     | job_title                    | min_salary | max_salary |
+------------+------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant            |       4200 |       9000 |
| AC_MGR     | Accounting Manager           |       8200 |      16000 |
| AD_ASST    | Administration Assistant     |       3000 |       6000 |
| AD_PRES    | President                    |      20000 |      40000 |
| AD_VP      | Administration Vice President |      15000 |      30000 |
| FI_ACCOUNT | Accountant                   |       4200 |       9000 |
| FI_MGR     | Finance Manager              |       8200 |      16000 |
| SA_MAN     | Sales Manager                |      10000 |      20000 |
+------------+------------------------------+------------+------------+
8 rows in set (0.00 sec)

mysql> UPDATE Employees
    ->   SET EMAIL = 'not available',
    -> COMMISSION_PCT = 0.10;
Query OK, 10 rows affected (0.06 sec)
Rows matched: 10  Changed: 10  Warnings: 0

mysql> select * from employees;
```

```
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |       200 |          10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|         0.10 |       200 |          10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |       200 |          10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|         0.10 |       103 |          60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |       103 |          60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |       114 |          30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |         0.10 |       145 |          80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |       145 |          80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

10 rows in set (0.00 sec)


mysql> select * from department;

+---------------+------------------+------------+-------------+

| Department_ID | Department_Name  | Manager_ID | Location_ID |

+---------------+------------------+------------+-------------+

|          10 | Administration   |       200 |       1700 |

|          20 | Marketing        |       201 |       1800 |
```

```
|          30 | Purchasing       |        114 |        1700 |
|          40 | Human Resources  |        203 |        2400 |
|          50 | Shipping         |        121 |        1500 |
|          60 | IT               |        103 |        1400 |
|          70 | Public Relations |        204 |        2700 |
|          80 | Sales            |        145 |        2500 |
|          90 | Finance          |        205 |        2700 |
|         100 | Legal            |        206 |        2200 |
|         110 | Customer Service |        207 |        2300 |
+---------------+------------------+------------+-------------+
11 rows in set (0.00 sec)

mysql> select * from jobs;
+------------+------------------------------+------------+------------+
| job_id     | job_title                    | min_salary | max_salary |
+------------+------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant            |       4200 |       9000 |
| AC_MGR     | Accounting Manager           |       8200 |      16000 |
| AD_ASST    | Administration Assistant     |       3000 |       6000 |
| AD_PRES    | President                    |      20000 |      40000 |
| AD_VP      | Administration Vice President |      15000 |      30000 |
| FI_ACCOUNT | Accountant                   |       4200 |       9000 |
| FI_MGR     | Finance Manager              |       8200 |      16000 |
| SA_MAN     | Sales Manager                |      10000 |      20000 |
+------------+------------------------------+------------+------------+
8 rows in set (0.00 sec)

mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
```

```
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |        200 |           10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|          0.10 |        200 |           10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |        200 |           10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|          0.10 |        103 |           60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |        103 |           60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select first_name,last_name from employee;

ERROR 1146 (42S02): Table 'assignment_dql_select.employee' doesn't exist

mysql> select first_name,last_name from employees;

+------------+-----------+

| first_name | last_name |

+------------+-----------+

| Steven     | King      |

| Neena      | Kochhar   |

| Lex        | De Haan   |
```

```
| Alexander  | Hunold    |
| Bruce      | Ernst     |
| David      | Austin    |
| Valli      | Pataballa |
| Diana      | Lorentz   |
| Nancy      | Greenberg |
| Daniel     | Faviet    |
+------------+-----------+
10 rows in set (0.00 sec)

mysql> select first_name as "First Name",last_name as "Last_name" from employee;
ERROR 1146 (42S02): Table 'assignment_dql_select.employee' doesn't exist
mysql> select first_name as "First Name",last_name as "Last_name" from employees;
+------------+-----------+
| First Name | Last_name |
+------------+-----------+
| Steven     | King      |
| Neena      | Kochhar   |
| Lex        | De Haan   |
| Alexander  | Hunold    |
| Bruce      | Ernst     |
| David      | Austin    |
| Valli      | Pataballa |
| Diana      | Lorentz   |
| Nancy      | Greenberg |
| Daniel     | Faviet    |
+------------+-----------+
10 rows in set (0.00 sec)

mysql> select * from employees;
```

```
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |       200 |          10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|         0.10 |       200 |          10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |       200 |          10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|         0.10 |       103 |          60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |       103 |          60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |       114 |          30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |         0.10 |       145 |          80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |       145 |          80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

10 rows in set (0.00 sec)


mysql> desc employees;

+----------------+---------------+------+-----+---------+-------+

| Field          | Type          | Null | Key | Default | Extra |

+----------------+---------------+------+-----+---------+-------+

| EMPLOYEE_ID    | int           | NO   | PRI | NULL    |       |

| FIRST_NAME     | varchar(50)   | YES  |     | NULL    |       |
```

```
| LAST_NAME      | varchar(50)   | YES  |     | NULL    |       |
| EMAIL          | varchar(50)   | YES  |     | NULL    |       |
| PHONE_NUMBER   | varchar(20)   | YES  |     | NULL    |       |
| HIRE_DATE      | date          | YES  |     | NULL    |       |
| JOB_ID         | varchar(20)   | YES  |     | NULL    |       |
| SALARY         | decimal(10,2) | YES  |     | NULL    |       |
| COMMISSION_PCT | decimal(5,2)  | YES  |     | NULL    |       |
| MANAGER_ID     | int           | YES  |     | NULL    |       |
| DEPARTMENT_ID  | int           | YES  |     | NULL    |       |
+----------------+---------------+------+-----+---------+-------+
11 rows in set (0.06 sec)

mysql> select * from wmployee
    -> where department_id = 80;
ERROR 1146 (42S02): Table 'assignment_dql_select.wmployee' doesn't exist
mysql> select * from employees
    -> where department_id = 80;
+-------------+------------+-----------+---------------+--------------+------------+--------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+--------+----------+----------------+-----------------+
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN | 13200.00 |           0.10 |        145 |            80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN |  9900.00 |           0.10 |        145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+--------+----------+----------------+-----------------+
2 rows in set (0.00 sec)

mysql> select * from employees
    -> where department_id = 30;
```

```
+-------------+------------+-----------+---------------+-------------+------------+---------+---------+----------------+-------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY  | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+-------------+------------+---------+---------+----------------+-------------+---------------+
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG | 4620.00 |           0.10 |         114 |            30 |
+-------------+------------+-----------+---------------+-------------+------------+---------+---------+----------------+-------------+---------------+
1 row in set (0.00 sec)


mysql> select * from employees
    -> order by name desc;
ERROR 1054 (42S22): Unknown column 'name' in 'order clause'
mysql> select * from employees
    -> order by first_name desc;
+-------------+------------+-----------+---------------+-------------+------------+---------+---------+----------------+-------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY  | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+-------------+------------+---------+---------+----------------+-------------+---------------+
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG | 5280.00 |           0.10 |         103 |            60 |
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |         200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  | 13200.00 |           0.10 |         145 |            80 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG | 4620.00 |           0.10 |         114 |            30 |
```

| 105 | David | Austin | not available | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 | 0.10 | 103 | 60 |

| 109 | Daniel | Faviet | not available | 515.124.4169 | 1987-06-26 | SA_MAN | 9900.00 | 0.10 | 145 | 80 |

| 104 | Bruce | Ernst | not available | 590.423.4568 | 1987-06-21 | IT_PROG | 6600.00 | 0.10 | 103 | 60 |

| 103 | Alexander | Hunold | not available | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 | 0.10 | 103 | 60 |

+----------+----------+----------+----------+----------+----------+----------+----------+----------+----------+

10 rows in set (0.00 sec)

mysql> select * from employees;

+----------+----------+----------+----------+----------+----------+----------+----------+----------+----------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+----------+----------+----------+----------+----------+----------+----------+----------+----------+----------+

| 100 | Steven | King | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 | 0.10 | 200 | 10 |

| 101 | Neena | Kochhar | not available | 515.123.4568 | 1987-06-18 | AD_VP | 18700.00 | 0.10 | 200 | 10 |

| 102 | Lex | De Haan | not available | 515.123.4569 | 1987-06-19 | AD_VP | 18700.00 | 0.10 | 200 | 10 |

| 103 | Alexander | Hunold | not available | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 | 0.10 | 103 | 60 |

| 104 | Bruce | Ernst | not available | 590.423.4568 | 1987-06-21 | IT_PROG | 6600.00 | 0.10 | 103 | 60 |

| 105 | David | Austin | not available | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 | 0.10 | 103 | 60 |

| 106 | Valli | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG | 5280.00 | 0.10 | 103 | 60 |

| 107 | Diana | Lorentz | not available | 590.423.5567 | 1987-06-24 | IT_PROG | 4620.00 | 0.10 | 114 | 30 |

| 108 | Nancy | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN | 13200.00 | 0.10 | 145 | 80 |

```
|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
10 rows in set (0.00 sec)


mysql> select first_name,last_name,salary from employees;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Nancy      | Greenberg | 13200.00 |
| Daniel     | Faviet    |  9900.00 |
+------------+-----------+----------+
10 rows in set (0.00 sec)


mysql> select first_name,last_name,salary,salary * 0.15 from employees;
+------------+-----------+----------+---------------+
| first_name | last_name | salary   | salary * 0.15 |
+------------+-----------+----------+---------------+
| Steven     | King      | 26400.00 |     3960.0000 |
| Neena      | Kochhar   | 18700.00 |     2805.0000 |
| Lex        | De Haan   | 18700.00 |     2805.0000 |
| Alexander  | Hunold    |  9900.00 |     1485.0000 |
| Bruce      | Ernst     |  6600.00 |      990.0000 |
```

```
| David      | Austin    |  5280.00 |      792.0000 |

| Valli      | Pataballa |  5280.00 |      792.0000 |

| Diana      | Lorentz   |  4620.00 |      693.0000 |

| Nancy      | Greenberg | 13200.00 |     1980.0000 |

| Daniel     | Faviet    |  9900.00 |     1485.0000 |

+------------+-----------+----------+---------------+

10 rows in set (0.00 sec)


mysql> select * from employees;

+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |       200 |           10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|          0.10 |       200 |           10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |       200 |           10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|          0.10 |       103 |           60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |       103 |           60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |       103 |           60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |       103 |           60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |       114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |       145 |           80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |       145 |           80 |
```

```
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
10 rows in set (0.00 sec)


mysql> select employee_id,first_name,last_name,salary from employees;
+-------------+------------+-----------+----------+
| employee_id | first_name | last_name | salary   |
+-------------+------------+-----------+----------+
|         100 | Steven     | King      | 26400.00 |
|         101 | Neena      | Kochhar   | 18700.00 |
|         102 | Lex        | De Haan   | 18700.00 |
|         103 | Alexander  | Hunold    |  9900.00 |
|         104 | Bruce      | Ernst     |  6600.00 |
|         105 | David      | Austin    |  5280.00 |
|         106 | Valli      | Pataballa |  5280.00 |
|         107 | Diana      | Lorentz   |  4620.00 |
|         108 | Nancy      | Greenberg | 13200.00 |
|         109 | Daniel     | Faviet    |  9900.00 |
+-------------+------------+-----------+----------+
10 rows in set (0.00 sec)


mysql> select * from employees
    -> order by salary;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |
```

```
|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |        103 |           60 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|          0.10 |        103 |           60 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|          0.10 |        200 |           10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |        200 |           10 |

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |        200 |           10 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+---------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select employee_id,first_name,last_name,salary from employee

    -> order by salary;

ERROR 1146 (42S02): Table 'assignment_dql_select.employee' doesn't exist

mysql> select employee_id,first_name,last_name,salary from employees

    -> order by salary;

+-------------+------------+-----------+----------+

| employee_id | first_name | last_name | salary   |

+-------------+------------+-----------+----------+

|        107 | Diana      | Lorentz   |  4620.00 |

|        105 | David      | Austin    |  5280.00 |

|        106 | Valli      | Pataballa |  5280.00 |

|        104 | Bruce      | Ernst     |  6600.00 |

|        103 | Alexander  | Hunold    |  9900.00 |

|        109 | Daniel     | Faviet    |  9900.00 |
```

```
|        108 | Nancy      | Greenberg | 13200.00 |
|        101 | Neena      | Kochhar   | 18700.00 |
|        102 | Lex        | De Haan   | 18700.00 |
|        100 | Steven     | King      | 26400.00 |
+-------------+------------+-----------+----------+
10 rows in set (0.00 sec)

mysql> select sum(salary) from employees;
+-------------+
| sum(salary) |
+-------------+
|   118580.00 |
+-------------+
1 row in set (0.06 sec)

mysql> select * from employees
    -> order by sum(salary);
ERROR 3029 (HY000): Expression #1 of ORDER BY contains aggregate function and applies to the result of a non-aggregated query
mysql> select sum(salary) as total salary from employees;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'salary from employees' at line 1
mysql> select sum(salary) as "total salaries" from employees;
+----------------+
| total salaries |
+----------------+
|      118580.00 |
+----------------+
1 row in set (0.00 sec)

mysql> select * from employees;
```

```
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |       200 |          10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|         0.10 |       200 |          10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |       200 |          10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|         0.10 |       103 |          60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |       103 |          60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |       114 |          30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |         0.10 |       145 |          80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |       145 |          80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+--------------+

10 rows in set (0.00 sec)


mysql> select max(salary) as "maximum salary",

    -> min(salary) as "minimum salary"

    -> from employees;

+----------------+----------------+

| maximum salary | minimum salary |

+----------------+----------------+
```

```
|      26400.00 |       4620.00 |
+----------------+----------------+
1 row in set (0.04 sec)

mysql> select * from employees
   -> order by avg(salary), count(employee_id);
ERROR 3029 (HY000): Expression #1 of ORDER BY contains aggregate function and applies to the result of a non-aggregated query
mysql> select avg(salary) as "average salary",
   -> count(employee_id) as total employees
   -> from employees;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'employees
from employees' at line 2
mysql> select avg(salary) as "average salary",
   -> count(employee_id) as "total employees"
   -> from employees;
+----------------+-----------------+
| average salary | total employees |
+----------------+-----------------+
|   11858.000000 |              10 |
+----------------+-----------------+
1 row in set (0.05 sec)

mysql> select count(*) as "total no. of employees working with the company"
   -> from employees);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')' at line 2
mysql> select count(*) as "total number of employees working with the company"
   -> from employees;
+--------------------------------------------------+
| total number of employees working with the company |
```

```
+-----------------------------------------------+
|                                        10 |
+-----------------------------------------------+
1 row in set (0.07 sec)


mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |        200 |           10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|          0.10 |        200 |           10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |        200 |           10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|          0.10 |        103 |           60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |        103 |           60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+
10 rows in set (0.00 sec)
```

```
mysql> select * from department;
+---------------+------------------+------------+-------------+
| Department_ID | Department_Name  | Manager_ID | Location_ID |
+---------------+------------------+------------+-------------+
|            10 | Administration   |        200 |        1700 |
|            20 | Marketing        |        201 |        1800 |
|            30 | Purchasing       |        114 |        1700 |
|            40 | Human Resources  |        203 |        2400 |
|            50 | Shipping         |        121 |        1500 |
|            60 | IT               |        103 |        1400 |
|            70 | Public Relations |        204 |        2700 |
|            80 | Sales            |        145 |        2500 |
|            90 | Finance          |        205 |        2700 |
|           100 | Legal            |        206 |        2200 |
|           110 | Customer Service |        207 |        2300 |
+---------------+------------------+------------+-------------+
11 rows in set (0.00 sec)

mysql> select * from jobs;
+------------+------------------------------+------------+------------+
| job_id     | job_title                    | min_salary | max_salary |
+------------+------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant            |       4200 |       9000 |
| AC_MGR     | Accounting Manager           |       8200 |      16000 |
| AD_ASST    | Administration Assistant     |       3000 |       6000 |
| AD_PRES    | President                    |      20000 |      40000 |
| AD_VP      | Administration Vice President |      15000 |      30000 |
| FI_ACCOUNT | Accountant                   |       4200 |       9000 |
| FI_MGR     | Finance Manager              |       8200 |      16000 |
| SA_MAN     | Sales Manager                |      10000 |      20000 |
+------------+------------------------------+------------+------------+
```

8 rows in set (0.00 sec)

mysql> select * from employees;

```
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |         200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |         103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |         103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |           0.10 |         103 |            60 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |           0.10 |         103 |            60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |           0.10 |         114 |            30 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  | 13200.00 |           0.10 |         145 |            80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |           0.10 |         145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
```

10 rows in set (0.00 sec)

mysql> select * from employees

-> order by job_id

-> limit 10;

```
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |       200 |          10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|         0.10 |       200 |          10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |       200 |          10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|         0.10 |       103 |          60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |       103 |          60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |       103 |          60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |       114 |          30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |         0.10 |       145 |          80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |       145 |          80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select count(distinct job_id) from employees;

+------------------------+

| count(distinct job_id) |

+------------------------+

|                      4 |

+------------------------+
```

1 row in set (0.00 sec)

```
mysql> select * from employees
    -> order by emp_id
    -> limit 1;
ERROR 1054 (42S22): Unknown column 'emp_id' in 'order clause'
mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |        200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |        103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |        103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |           0.10 |        103 |            60 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |           0.10 |        103 |            60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |           0.10 |        114 |            30 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  | 13200.00 |           0.10 |        145 |            80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |           0.10 |        145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
10 rows in set (0.00 sec)
```

```
mysql> select * from employees
    -> order by employee_id
    -> limit 1;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |         200 |            10 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
1 row in set (0.00 sec)


mysql> select * from employees
    -> order by employee_id
    -> limit 10;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |         200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |         103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |         103 |            60 |
```

```
|        105 | David      | Austin     | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        107 | Diana      | Lorentz    | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        109 | Daniel     | Faviet     | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select first_name,last_name,salary from employees

    -> where salary not in between 10000 and 15000;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
your MySQL server version for the right syntax to use near 'between 10000 and 15000' at line 2

mysql> select first_name,last_name,salary from employees

    -> where salary not between 10000 and 15000;

+------------+-----------+----------+

| first_name | last_name | salary   |

+------------+-----------+----------+

| Steven     | King      | 26400.00 |

| Neena      | Kochhar   | 18700.00 |

| Lex        | De Haan   | 18700.00 |

| Alexander  | Hunold    |  9900.00 |

| Bruce      | Ernst     |  6600.00 |

| David      | Austin    |  5280.00 |

| Valli      | Pataballa |  5280.00 |

| Diana      | Lorentz   |  4620.00 |

| Daniel     | Faviet    |  9900.00 |

+------------+-----------+----------+
```

```
9 rows in set (0.05 sec)

mysql> select first_name,last_name,salary from employees
    -> order by salary
    -> where salary not between 10000 and 15000;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'where salary not between 10000 and 15000' at line 3
mysql> select first_name,last_name,salary from employees
    -> where salary < 10000 and salary > 15000;
Empty set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary < 10000 or salary > 15000;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Daniel     | Faviet    |  9900.00 |
+------------+-----------+----------+
9 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000
    -> order by first_name,last_name,salary;
```

```
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| Daniel     | Faviet    |  9900.00 |
| David      | Austin    |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Lex        | De Haan   | 18700.00 |
| Neena      | Kochhar   | 18700.00 |
| Steven     | King      | 26400.00 |
| Valli      | Pataballa |  5280.00 |
+------------+-----------+----------+
9 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000
    -> order by salary desc;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Daniel     | Faviet    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
+------------+-----------+----------+
```

9 rows in set (0.00 sec)

```
mysql> select * from employees
    -> where salary not between 10000 and 15000
    -> order by first_name,last_name,salary;
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----------------+
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 |           0.10 |        103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG | 6600.00 |           0.10 |        103 |            60 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  | 9900.00 |           0.10 |        145 |            80 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 |           0.10 |        103 |            60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG | 4620.00 |           0.10 |        114 |            30 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |        200 |            10 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG | 5280.00 |           0.10 |        103 |            60 |
+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----------------+
9 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000;
+------------+-----------+----------+
```

```
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Daniel     | Faviet    |  9900.00 |
+------------+-----------+----------+
9 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000
    -> order by salary;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Diana      | Lorentz   |  4620.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Bruce      | Ernst     |  6600.00 |
| Alexander  | Hunold    |  9900.00 |
| Daniel     | Faviet    |  9900.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Steven     | King      | 26400.00 |
+------------+-----------+----------+
9 rows in set (0.00 sec)
```

```
mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000
    -> order by first_name,last_name,salary;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| Daniel     | Faviet    |  9900.00 |
| David      | Austin    |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Lex        | De Haan   | 18700.00 |
| Neena      | Kochhar   | 18700.00 |
| Steven     | King      | 26400.00 |
| Valli      | Pataballa |  5280.00 |
+------------+-----------+----------+
9 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salary not between 10000 and 15000;
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
```

```
| Diana      | Lorentz   |  4620.00 |

| Daniel     | Faviet    |  9900.00 |

+------------+-----------+----------+

9 rows in set (0.00 sec)


mysql> select * from employees;

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |        200 |           10 |

|        101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00
|          0.10 |        200 |           10 |

|        102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |
0.10 |        200 |           10 |

|        103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00
|          0.10 |        103 |           60 |

|        104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |
0.10 |        103 |           60 |

|        105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)
```

```
mysql> select first_name,last_name,department_id from employees
    -> where department_id between 30 and 100;
+------------+-----------+---------------+
| first_name | last_name | department_id |
+------------+-----------+---------------+
| Alexander  | Hunold    |            60 |
| Bruce      | Ernst     |            60 |
| David      | Austin    |            60 |
| Valli      | Pataballa |            60 |
| Diana      | Lorentz   |            30 |
| Nancy      | Greenberg |            80 |
| Daniel     | Faviet    |            80 |
+------------+-----------+---------------+
7 rows in set (0.00 sec)

mysql> select first_name,last_name,department_id from employees
    -> where department_id between 30 and 100
    -> order by department_id;
+------------+-----------+---------------+
| first_name | last_name | department_id |
+------------+-----------+---------------+
| Diana      | Lorentz   |            30 |
| Alexander  | Hunold    |            60 |
| Bruce      | Ernst     |            60 |
| David      | Austin    |            60 |
| Valli      | Pataballa |            60 |
| Nancy      | Greenberg |            80 |
| Daniel     | Faviet    |            80 |
+------------+-----------+---------------+
7 rows in set (0.00 sec)
```

```
mysql> select select * from employees
    -> where department_id between 30 and 100;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'select * from employees
where department_id between 30 and 100' at line 1
mysql> select * from employees
    -> where department_id between 30 and 100;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |        103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |        103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |           0.10 |        103 |            60 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |           0.10 |        103 |            60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |           0.10 |        114 |            30 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  | 13200.00 |           0.10 |        145 |            80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |           0.10 |        145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
7 rows in set (0.00 sec)


mysql> select first_name,last_name,department_id from employees
    -> where department_id between 30 and 100;
+------------+-----------+---------------+
| first_name | last_name | department_id |
```

```
+------------+-----------+---------------+
| Alexander  | Hunold    |            60 |
| Bruce      | Ernst     |            60 |
| David      | Austin    |            60 |
| Valli      | Pataballa |            60 |
| Diana      | Lorentz   |            30 |
| Nancy      | Greenberg |            80 |
| Daniel     | Faviet    |            80 |
+------------+-----------+---------------+
7 rows in set (0.00 sec)

mysql> select first_name,last_name,department_id from employees
    -> where department_id in(30,100)
    -> order by department_id asc;
+------------+-----------+---------------+
| first_name | last_name | department_id |
+------------+-----------+---------------+
| Diana      | Lorentz   |            30 |
+------------+-----------+---------------+
1 row in set (0.00 sec)

mysql> select first_name,last_name,department_id from employees
    -> where department_id in(30,100)
    -> order by department_id;
+------------+-----------+---------------+
| first_name | last_name | department_id |
+------------+-----------+---------------+
| Diana      | Lorentz   |            30 |
+------------+-----------+---------------+
1 row in set (0.00 sec)
```

```
mysql> select first_name,last_name,department_id from employees
    -> where department_id between 30 and 100
    -> order department_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'department_id' at line 3
mysql> select first_name,last_name,department_id from employees
    -> where department_id between 30 and 100
    -> order by department_id;
+------------+-----------+---------------+
| first_name | last_name | department_id |
+------------+-----------+---------------+
| Diana      | Lorentz   |            30 |
| Alexander  | Hunold    |            60 |
| Bruce      | Ernst     |            60 |
| David      | Austin    |            60 |
| Valli      | Pataballa |            60 |
| Nancy      | Greenberg |            80 |
| Daniel     | Faviet    |            80 |
+------------+-----------+---------------+
7 rows in set (0.00 sec)


mysql> select first_name,last_name,salary from employees
    -> where salary not in (10000,15000), department_id between 30 and 100;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ', department_id between 30 and 100' at line 2
mysql> select first_name,last_name,salary from employees
    -> where salary not in (10000,15000);
+------------+-----------+----------+
| first_name | last_name | salary   |
+------------+-----------+----------+
| Steven     | King      | 26400.00 |
```

```
| Neena      | Kochhar   | 18700.00 |
| Lex        | De Haan   | 18700.00 |
| Alexander  | Hunold    |  9900.00 |
| Bruce      | Ernst     |  6600.00 |
| David      | Austin    |  5280.00 |
| Valli      | Pataballa |  5280.00 |
| Diana      | Lorentz   |  4620.00 |
| Nancy      | Greenberg | 13200.00 |
| Daniel     | Faviet    |  9900.00 |
+------------+-----------+----------+
10 rows in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> department_id in (30,100);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'in (30,100)' at line 2
mysql> select first_name,last_name,salary from employees
    -> where department_id in (30,100);
+------------+-----------+---------+
| first_name | last_name | salary  |
+------------+-----------+---------+
| Diana      | Lorentz   | 4620.00 |
+------------+-----------+---------+
1 row in set (0.00 sec)

mysql> select first_name,last_name,salary from employees
    -> where salar
    ->
    -> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')' at line 4
mysql> select first_name,last_name,salary from employees
```

```
    -> where salary not between 10000 and 15000
    -> and department_id in (30,100);
+------------+-----------+---------+
| first_name | last_name | salary  |
+------------+-----------+---------+
| Diana      | Lorentz   | 4620.00 |
+------------+-----------+---------+
1 row in set (0.00 sec)

mysql> select first_name,last_name,hire_date from employees
    -> where date like "1987%";
ERROR 1054 (42S22): Unknown column 'date' in 'where clause'
mysql> select first_name,last_name,hire_date from employees
    -> where hire_date like "1987%";
+------------+-----------+------------+
| first_name | last_name | hire_date  |
+------------+-----------+------------+
| Steven     | King      | 1987-06-17 |
| Neena      | Kochhar   | 1987-06-18 |
| Lex        | De Haan   | 1987-06-19 |
| Alexander  | Hunold    | 1987-06-20 |
| Bruce      | Ernst     | 1987-06-21 |
| David      | Austin    | 1987-06-22 |
| Valli      | Pataballa | 1987-06-23 |
| Diana      | Lorentz   | 1987-06-24 |
| Nancy      | Greenberg | 1987-06-25 |
| Daniel     | Faviet    | 1987-06-26 |
+------------+-----------+------------+
10 rows in set (0.05 sec)

mysql> select first_name from employees
```

```
    -> where first_name = "b" and first_name = "c";
Empty set (0.05 sec)

mysql> select first_name from employees
    -> where first_name = "b" or first_name = "c";
Empty set (0.00 sec)

mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |         200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |         200 |            10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |         103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |         103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |           0.10 |         103 |            60 |
|         106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |           0.10 |         103 |            60 |
|         107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |           0.10 |         114 |            30 |
|         108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  | 13200.00 |           0.10 |         145 |            80 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |           0.10 |         145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----------------+
```

```
10 rows in set (0.00 sec)

mysql> select first_name from employees
    -> where first_name like "%b%"
    -> and first_name like "%c%";
+------------+
| first_name |
+------------+
| Bruce      |
+------------+
1 row in set (0.00 sec)

mysql> where first_name like "%b%"
    -> or first_name like "%c%";
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'where first_name like "%b%"
or first_name like "%c%"' at line 1
mysql> select first_name from employees
    -> where first_name like "%b%"
    -> or first_name like "%c%";
+------------+
| first_name |
+------------+
| Bruce      |
| Nancy      |
+------------+
2 rows in set (0.00 sec)

mysql> select last_name,job_id,salary from employees;
+-----------+---------+----------+
| last_name | job_id  | salary   |
```

```
+-----------+---------+----------+
| King      | AD_PRES | 26400.00 |
| Kochhar   | AD_VP   | 18700.00 |
| De Haan   | AD_VP   | 18700.00 |
| Hunold    | IT_PROG |  9900.00 |
| Ernst     | IT_PROG |  6600.00 |
| Austin    | IT_PROG |  5280.00 |
| Pataballa | IT_PROG |  5280.00 |
| Lorentz   | IT_PROG |  4620.00 |
| Greenberg | SA_MAN  | 13200.00 |
| Faviet    | SA_MAN  |  9900.00 |
+-----------+---------+----------+
10 rows in set (0.00 sec)

mysql> select * from employees;
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+---------------+
|         100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |           0.10 |        200 |            10 |
|         101 | Neena      | Kochhar   | not available | 515.123.4568 | 1987-06-18 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         102 | Lex        | De Haan   | not available | 515.123.4569 | 1987-06-19 | AD_VP   | 18700.00 |           0.10 |        200 |            10 |
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG |  9900.00 |           0.10 |        103 |            60 |
|         104 | Bruce      | Ernst     | not available | 590.423.4568 | 1987-06-21 | IT_PROG |  6600.00 |           0.10 |        103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG |  5280.00 |           0.10 |        103 |            60 |
```

```
|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select * from department;

+---------------+------------------+------------+-------------+

| Department_ID | Department_Name  | Manager_ID | Location_ID |

+---------------+------------------+------------+-------------+

|           10 | Administration   |        200 |        1700 |

|           20 | Marketing        |        201 |        1800 |

|           30 | Purchasing       |        114 |        1700 |

|           40 | Human Resources  |        203 |        2400 |

|           50 | Shipping         |        121 |        1500 |

|           60 | IT               |        103 |        1400 |

|           70 | Public Relations |        204 |        2700 |

|           80 | Sales            |        145 |        2500 |

|           90 | Finance          |        205 |        2700 |

|          100 | Legal            |        206 |        2200 |

|          110 | Customer Service |        207 |        2300 |

+---------------+------------------+------------+-------------+

11 rows in set (0.00 sec)


mysql> select * from jobs;

+------------+-------------------------------+------------+------------+

| job_id     | job_title                     | min_salary | max_salary |
```

```
+------------+------------------------------+------------+------------+
| AC_ACCOUNT | Public Accountant            |       4200 |       9000 |
| AC_MGR     | Accounting Manager           |       8200 |      16000 |
| AD_ASST    | Administration Assistant     |       3000 |       6000 |
| AD_PRES    | President                    |      20000 |      40000 |
| AD_VP      | Administration Vice President |      15000 |      30000 |
| FI_ACCOUNT | Accountant                   |       4200 |       9000 |
| FI_MGR     | Finance Manager              |       8200 |      16000 |
| SA_MAN     | Sales Manager                |      10000 |      20000 |
+------------+------------------------------+------------+------------+
8 rows in set (0.00 sec)

mysql> select last_name,job_id,salary from employees
    -> where job in ("Programmer","Shipping Clerk")
    -> and salary not in (4500,10000,15000);
ERROR 1054 (42S22): Unknown column 'job' in 'where clause'
mysql> select last_name,job_id,salary from employees
    -> where job_id in ("Programmer","Shipping Clerk")
    -> and salary not in (4500,10000,15000);
Empty set (0.00 sec)

mysql> select * from employees
    -> where len(first_name,last_name) = 6;
ERROR 1305 (42000): FUNCTION assignment_dql_select.len does not exist
mysql>
mysql> select * from employees
    -> where len(first_name) = 6 or len(last_name) = 6;
ERROR 1305 (42000): FUNCTION assignment_dql_select.len does not exist
mysql> select last_name from employees
    -> where length(last_name) = 6;
+-----------+
```

```
| last_name |
+-----------+
| Hunold    |
| Austin    |
| Faviet    |
+-----------+
3 rows in set (0.00 sec)

mysql> select * from employees
    -> where length(last_name) = 6;
+-------------+------------+-----------+---------------+--------------+------------+---------+---------+----------------+-------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL         | PHONE_NUMBER | HIRE_DATE  | JOB_ID  | SALARY  | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------------+--------------+------------+---------+---------+----------------+-------------+---------------+
|         103 | Alexander  | Hunold    | not available | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 |           0.10 |         103 |            60 |
|         105 | David      | Austin    | not available | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 |           0.10 |         103 |            60 |
|         109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  | 9900.00 |           0.10 |         145 |            80 |
+-------------+------------+-----------+---------------+--------------+------------+---------+---------+----------------+-------------+---------------+
3 rows in set (0.00 sec)

mysql> select last_name from employees
    -> where last_name like "__e%";
+-----------+
| last_name |
+-----------+
| Greenberg |
+-----------+
```

1 row in set (0.00 sec)

mysql> select * from employees

-> where last_name like "__e%";

+----------+----------+----------+----------+----------+----------+--------+----------+----------+----------+----------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+----------+----------+----------+----------+----------+----------+--------+----------+----------+----------+----------+

| 108 | Nancy | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN | 13200.00 | 0.10 | 145 | 80 |

+----------+----------+----------+----------+----------+----------+--------+----------+----------+----------+----------+

1 row in set (0.00 sec)

mysql> select * from employees;

+----------+----------+----------+----------+----------+----------+--------+----------+----------+----------+----------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+----------+----------+----------+----------+----------+----------+--------+----------+----------+----------+----------+

| 100 | Steven | King | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 | 0.10 | 200 | 10 |

| 101 | Neena | Kochhar | not available | 515.123.4568 | 1987-06-18 | AD_VP | 18700.00 | 0.10 | 200 | 10 |

| 102 | Lex | De Haan | not available | 515.123.4569 | 1987-06-19 | AD_VP | 18700.00 | 0.10 | 200 | 10 |

| 103 | Alexander | Hunold | not available | 590.423.4567 | 1987-06-20 | IT_PROG | 9900.00 | 0.10 | 103 | 60 |

| 104 | Bruce | Ernst | not available | 590.423.4568 | 1987-06-21 | IT_PROG | 6600.00 | 0.10 | 103 | 60 |

| 105 | David | Austin | not available | 590.423.4569 | 1987-06-22 | IT_PROG | 5280.00 | 0.10 | 103 | 60 |

```
|        106 | Valli      | Pataballa | not available | 590.423.4560 | 1987-06-23 | IT_PROG |  5280.00 |
0.10 |        103 |           60 |

|        107 | Diana      | Lorentz   | not available | 590.423.5567 | 1987-06-24 | IT_PROG |  4620.00 |
0.10 |        114 |           30 |

|        108 | Nancy      | Greenberg | not available | 515.124.4569 | 1987-06-25 | SA_MAN  |
13200.00 |          0.10 |        145 |           80 |

|        109 | Daniel     | Faviet    | not available | 515.124.4169 | 1987-06-26 | SA_MAN  |  9900.00 |
0.10 |        145 |           80 |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

10 rows in set (0.00 sec)


mysql> select distinct job_id from employees;

+---------+

| job_id  |

+---------+

| AD_PRES |

| AD_VP   |

| IT_PROG |

| SA_MAN  |

+---------+

4 rows in set (0.00 sec)


mysql> select * from employees

    -> where last_name in("blake","scott","king","ford");

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL        | PHONE_NUMBER | HIRE_DATE  |
JOB_ID  | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |

+-------------+------------+-----------+---------------+-------------+------------+---------+----------+----------------+-----
-------+---------------+

|        100 | Steven     | King      | not available | 515.123.4567 | 1987-06-17 | AD_PRES | 26400.00 |
0.10 |        200 |           10 |
```

```
+-------------+------------+-----------+---------------+--------------+------------+---------+----------+----------------+-------
-------+---------------+
```

1 row in set (0.00 sec)

mysql>